

Owen Wu

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EDUCATION

University of Waterloo

Bachelor of Computer Science

Waterloo, ON

September 2026 – Present

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, Java, C++, SQL

Frameworks: Next.js, React, Django, Flask, Tailwind CSS, discord.js

Backend: RESTful APIs, MongoDB, Mongoose, Google Cloud Platform

Developer Tools: Git, GitHub, Jira, Docker, Vercel, npm, Yarn, Power BI, Power Apps, Power Automate, L^AT_EX

PROJECTS

StarWeave | *TypeScript, Tailwind CSS, RESTful API Design*

May 2026 - Present

- Built a full-stack celestial object database that integrates NASA Exoplanet Archive and JPL Horizons data using Next.js API integration, Tailwind CSS, and TypeScript, aggregating 40K+ astronomical objects
- Designed a server-side data pipeline and RESTful APIs using Next.js API integration to retrieve and search astronomical datasets, improving performance by 15% from 2000ms to 1700ms through RegEx-based filtering
- Developed reusable TypeScript interfaces and utility modules to standardize data from multiple NASA APIs into a unified format, simplifying frontend integration and supporting future expansion to additional data sources

DMOJ Open-Source Contributor | *Python, Django, SQL, Git*

May 2026 - Present

- Implemented dynamic username styling logic to ensure user search results accurately reflected their rank-based color assignments using CSS and Django templates, improving visual consistency and user experience on the site
- Developing a Discord-inspired username system that introduces immutable user identifiers using Python, Django, and SQL, enabling username changes while preserving username references through the application

Graph Theory Algorithm Visualizer | *Java, Processing, OOP*

December 2025 - January 2026

- Built an interactive graph theory visualizer using Java and Processing, allowing users to create, modify, and visualize custom mathematical graph representations through a dynamic graphical interface
- Implemented Breadth-First Search (BFS) and Depth-First Search (DFS) algorithms with real-time animations using object-oriented design, providing step-by-step visualization of graph traversal for ease of understanding

EXPERIENCE

Software Engineer Intern

July 2025 – August 2025

Nokia

- Developed a centralized document catalogue application managing 600+ assets for the Data Analyst team using Microsoft Power BI and Power Apps, improving documentation efficiency and accessibility
- Engineered API workflows that synchronized Excel-based reporting data and automatically distributed stakeholder notifications when an asset was updated using Microsoft Power Automate, reducing manual operational overhead
- Collaborated with a 10-member Data Analysis team to identify and implement performance improvements in their Power BI reports through Agile development practices, iterative development, and testing

Full-stack Developer

August 2025 – June 2026

JAMHacks

- Developed the Sponsors and Meet the Team sections of the JamHacks website using Next.js and Tailwind CSS, contributing to a production platform that served 10K+ website impressions
- Built RESTful APIs with Next.js API routes and Mongoose to automate applicant acceptance, rejection, and waitlist workflows, improving the process by removing the need for manual applicant processing
- Built a location lookup command and mentor ticketing system with Discord.js and Google Cloud Platform, enabling 250+ hackers to find venue locations and connect with specialized mentors more efficiently
- Integrated an API that summarized all of the participant data using MongoDB and Google Vertex AI API for a 19-member organizing team, increasing hacker satisfaction through additional merchandise and events

AWARDS AND ACHIEVEMENTS

Canadian Computing Contest Senior: Used Python, Java, and C++ to solve programming problems that cover topics from mathematics to various data structures and algorithms. Top 25% of participants in 2024, 2025, and 2026